Project Proposal: Agentic Tango DJ (@DJ)

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1. Problem Statement & Objectives

Problem Statement

Traditional Tango events (*Milongas*) follow a rigid structure of *tandas* (3-4 songs of one genre) separated by *cortinas* (non-tango breaks). Currently, events without human DJs rely on either automated static playlists or manual selection by dancers. Both approaches fail: static playlists cannot react to the room's energy, and manual selection is impractical because most dancers seek a specific mood rather than memorizing extensive song catalogs. To date, no agentic AI system translates these semantic requests into a dynamically managed Milonga framework while incorporating real-time feedback.

Objectives

AT-DJ is an agentic system ensuring traditional Milonga compliance while offering live DJ sensitivity. Outcomes include:

- **Dynamic Adaptation:** Generating valid tandas and adjusting tracks via "almost-live" user feedback.
- **Audio Empowerment:** Automatically enhancing historical Tango track audio quality.
- **Interactive Q&A:** Retrieving track metadata to answer user queries.
- **Generative Transitions:** Synthesizing *cortinas* based on preceding tanda features for seamless blending.

Target Deliverables

A documented GitHub repository and an interactive web app demonstrating real-time adaptability.

Alignment with Course Objectives

This project practically implements core course topics by explicitly utilizing tool-assisted LLMs, autonomous agentic systems, and RAG.

2. Technical Approach & Timeline

We will deliver an end-to-end prototype within the 5-week timeframe:

Timeline & Milestones

- **Phase 1: Data Pipeline & RAG Setup (Weeks 1-2):** Extract music features using audio analysis libraries (e.g., Essentia, Librosa). Construct a vector database for semantic search over audio features.
- **Phase 2: Agent & Tool Integration (Weeks 3-4):** Develop the reasoning engine using an agentic framework (e.g., LangGraph) and an LLM. Implement specialized tools: an audio enhancement module, a transition music synthesizing tool, and a music QA retrieval tool.
- **Phase 3: UI Integration & Evaluation (Week 5):** Build an interactive web application (e.g., Streamlit) to manage the live queue and user feedback loop. Evaluate system latency and semantic accuracy.

Resource Assessment & Availability

To manage limited computational and API budgets, heavy audio feature extraction will run locally offline. For reasoning, we will efficiently route calls to cost-effective models, using cloud resources strictly as needed.

Potential Challenges & Mitigations

- **Generative Audio Latency:** Generation risk dead air. *Mitigation:* Asynchronously trigger generation during the tanda's final song. Fallback: retrieve and crossfade pre-existing non-tango tracks matching transition features.
- **Milonga Constraints:** LLMs may hallucinate invalid setlists. *Mitigation:* Embed hard-coded logic guardrails within the agent framework to ensure structural compliance before queuing.

3. Innovation & Relevance

Unlike general AI DJs designed for unstructured solo listening, AT-DJ adapts Agentic AI to a strict, communal cultural event. Traditional recommendation algorithms cannot manage Tango's rigid *tandas*/*cortinas* structure. We bridge GenAI trends with this domain via tool-assisted reasoning and RAG. The agent actively reasons about dancer energy and autonomously invokes external tools. This "sets, breaks, and energy flow" logic establishes a framework easily generalizable to other structured social dances (e.g., Salsa, Swing).